

A Randomized Controlled Trial of the Safety and Effectiveness of a Contact Force Sensing Irrigated Catheter for Ablation of Paroxysmal Atrial Fibrillation: Results of the TOCCASTAR Study

Running title: Reddy *et al.*; Results of contact force sensing TOCCASTAR study

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Abstract

Background—Contact Force (CF) is a major determinant of lesion size and transmurality, and has the potential to improve efficacy of atrial fibrillation (AF) ablation. This study sought to evaluate the safety and effectiveness of a novel irrigated radiofrequency ablation (RFA) catheter which measures real-time CF in the treatment of patients with paroxysmal AF.

Methods and Results—A total of 300 patients with symptomatic, drug-refractory, paroxysmal AF were enrolled in a prospective, multicenter, randomized controlled trial and randomized to RFA with either a novel CF sensing catheter or a non-CF catheter (Control). The primary effectiveness endpoint consisted of acute electrical isolation of all pulmonary veins (PVs) and freedom from recurrent symptomatic atrial arrhythmia off all anti-arrhythmic drugs at 12 months. The primary safety endpoint included device-related serious adverse events. Endpoints were powered to show non-inferiority. All PVs were isolated in both groups. Effectiveness was achieved in 67.8% and 69.4% of subjects in the CF and Control arms, respectively (absolute difference -1.6%, lower limit of one-sided 95% CI -10.7%, $p=0.0073$ for non-inferiority). When the CF arm was stratified into optimal CF ($\geq 90\%$ ablations with $\geq 10\text{g}$) and non-optimal CF groups, effectiveness was achieved in 75.9% vs. 58.1%, respectively ($p=0.018$). The primary safety endpoint occurred in 1.97% and 1.40% of CF patients and Controls, respectively (absolute difference 0.57%, upper limit of one-sided 95% CI 3.61%, $p=0.0004$ for non-inferiority).

Conclusions—The CF ablation catheter met the primary safety and effectiveness endpoints. Additionally, optimal CF was associated with improved effectiveness.

Clinical Trial Registration Information—clinicaltrials.gov; Identifier: NCT01278953.

Key words: radiofrequency ablation, atrial fibrillation, contact force, catheter ablation

Introduction

Multiple randomized clinical trials have demonstrated that catheter ablation is superior to anti-arrhythmic medications for the treatment of paroxysmal atrial fibrillation (AF).¹⁻⁴ While conceptually straightforward, the placement of continuous point-by-point radiofrequency (RF) ablation lesions around the pulmonary veins (PVs) is technically challenging. Achieving a continuous ring of atrial necrosis without conduction gaps can be difficult. Although acute electrical PV isolation is almost universally achieved eventually, the mechanism of acute isolation is typically a combination of myocardial necrosis but also a significant degree of tissue edema often resulting in reversible injury.⁵ With resolution of reversible injury, PV reconnections occur and may again result in AF. Indeed, long-term follow-up studies of AF patients who underwent PV isolation show continued attrition of efficacy over time, predominantly due to PV reconnections in the vast majority.⁶⁻⁸ Durable PV Isolation can be an elusive goal.

Technical limitations in point-by-point RF energy delivery have prompted intense development to improve the tools employed to achieve durable PV isolation. A limitation of standard RF ablation has been the inability to directly assess the degree and quality of contact between the catheter tip and target tissue prompting operators to rely on surrogate markers of contact such as baseline impedance, impedance change, electrode temperature, and catheter location by fluoroscopy or electroanatomical mapping.⁹⁻¹¹ Lesion formation depends on a combination of, RF power, duration and contact force (CF), but CF has not been measurable thus far. Recently, ablation catheters have been introduced that have the capability of quantifying the CF between the catheter tip and tissue in real time¹²⁻¹⁶. In pre-clinical studies, a direct relationship between the CF measured by these catheters and lesion size independent of

delivered RF power has been established.^{14, 15, 17} In addition, a lesion with poor CF may lead to edema without a durable lesion but make the creation of a durable lesion more difficult.

Furthermore, several nonrandomized clinical studies have demonstrated the potential benefit of knowing and applying CF information during AF ablation.^{12, 13, 16} These studies have set the stage for a multicenter, prospective, randomized controlled trial comparing a CF sensing catheter with a standard catheter with respect to safety and efficacy.

Methods

The TOCCASTAR Study (TactiCath Contact Force Ablation Catheter Study for Atrial Fibrillation) was a prospective, randomized controlled, multicenter study conducted to evaluate the safety and effectiveness of the CF catheter (TactiCath™, St. Jude Medical, St. Paul, MN, USA) for the treatment of symptomatic paroxysmal AF. The study was conducted under an IDE for regulatory approval and was designed as a non-inferiority trial for safety and effectiveness. It was not powered for superiority. Subjects were enrolled at 17 clinical sites and treated by 47 operators in Europe and the US. The Ethics Committee or Institutional Review Board of each participating institution approved the study protocol and informed written consent was obtained from all patients. An independent Data Safety Monitoring Board supervised the study.

Patient Selection

Patients with symptomatic, paroxysmal AF that was refractory or intolerant to at least one class I-IV anti-arrhythmic drug (AAD) who were planned for catheter ablation were eligible for participation in the study. Subjects were required to have at least one documented episode of AF > 30 seconds and a minimum of three episodes documented by history within the 12 months prior to enrollment. Key exclusion criteria included persistent AF, myocardial infarction within

the three months prior to the intervention, prior left atrial ablation (surgical or catheter), left atrial diameter > 5 cm, left ventricular ejection fraction < 35%, contraindications to long-term antithrombotic therapy, and severe pulmonary disease.

Treatment Arms

Patients were randomized in a 1:1 ratio to catheter ablation with either the TactiCath CF catheter or Control catheter (ThermoCool Navistar™, Biosense Webster, Diamond Bar, CA, USA). Both catheters are dimensionally equivalent and were designed with 6 holes for open irrigated RF ablation. They also have a multi-electrode design with deflectable tip, allowing monitoring of intra-cardiac signals and pacing capabilities. The distinguishing feature of the CF catheter is the three fiber optic sensors that are located at an angle of 120 degrees from each other within the tip. These sensors measure micro-deformations which correlate to the applied CF between the catheter tip electrode and the underlying tissue.^{14, 17} These small fiber optic lines allow force to be measured very close to the tip. The TactiSys™ system (St. Jude Medical, St. Paul, MN, USA) displays detailed CF information by determining the cross product of axial and lateral force components to obtain the resultant force vector, fully describing the force at the point of contact between the catheter and tissue surface. The system displays the calculated force time integral (FTI) which is determined using the CF and the duration of RF application during ablation.¹⁵ The 3D electroanatomical mapping system used was dictated by the labeling of each device. For subjects randomized to the CF catheter, the EnSite™ NavX™ electroanatomic mapping system (St. Jude Medical, St. Paul, MN, USA) was used. For Control subjects undergoing ablation with the control catheter, the CARTO® electroanatomical mapping system (Biosense Webster, Diamond Bar, CA, USA) was used.

To allow for initial familiarization with the CF device, up to two subjects per site were

enrolled as roll-in patients.

Catheter Ablation Procedure

The index procedure involved isolation of all PVs performed according to the physician's standard practice. Acute electrical PV isolation was confirmed by the presence of entrance block 30 minutes after complete isolation of all PVs. Additionally, isoproterenol-induced latent PV conduction and non-PV triggers were identified and ablated according to the physician's discretion. Additional lines and CFAEs were targeted at physician discretion. Ablations were performed with the minimum RF power deemed necessary by the operator and with a standardized, power-dependent irrigation rate. When using the study device, the CF and derived parameters were provided to the operator in real-time and recorded throughout the procedure.

Importantly, no target CF guidelines were specified by the protocol and CF application during this study was based on clinical data from other studies that became available as this study progressed. The protocol allowed additional left atrial ablations to treat known or suspected arrhythmia triggers provided that the catheter assigned by randomization was used. A right atrial isthmus line ablation was recommended if atrial flutter was documented either prior to or during the procedure.

Follow-Up

Patients were followed for 12 months after the index procedure and evaluated with office visits at 7 days, 3, 6 and 12 months after the initial ablation. During the 3-month blanking period following the index procedure, a maximum of 2 repeat ablation procedures were allowed using the catheter type assigned at the index procedure. During the blanking period, the use of AADs was allowed, but all class I and III AADs were to be discontinued at the end of the blanking period.

Transtelephonic monitoring (TTM) was required after the blanking period with weekly transmission until 5 months after ablation followed by monthly transmissions. Subjects transmitted additional unscheduled TTMs anytime symptoms recurred. Ambulatory 24h Holter monitoring was performed at the end of the study. All TTMs were evaluated by an independent core-lab.

Study Endpoints

The primary effectiveness endpoint reflected the combined acute and chronic ablation success. Acute ablation success was defined as complete isolation of all PVs at the end of the index procedure. Chronic ablation success was defined as freedom from recurrence of symptomatic atrial fibrillation, atrial tachycardia, or atrial flutter at 12 months after the index procedure off AADs, with the exclusion of the 3 month blanking period. Any acute procedural failure, retreatment with ablation, use of class I or III AAD, or symptomatic recurrence documented by TTM or ECG following the blanking period constituted treatment failure. Unlike other recently reported studies, use of any class I or III AAD beyond the 3 month blanking period was considered a failure even if no AF was documented.^{3, 13}

A ‘worst case’ sensitivity analysis was performed on the ‘modified Intension-To-Treat (mITT)’ population. The mITT population included all randomized patients in whom a study device was introduced and PVI was attempted. The ‘worst case’ situation imputed failures for subjects who terminated early without failure from the Study arm and imputed successes for subjects who terminated early without failure from the Control arm.

An alternate definition of treatment success, referred to as ‘clinically relevant effectiveness’, was also used to provide a more clinically meaningful measure of effectiveness. The differences between primary and ‘clinically relevant’ effectiveness failure consist of 1)

failure due to AAD use post blanking needs to be associated with documentation of arrhythmia through TTM or ECG, and 2) failure due to TTM only when at least three positive TTMs are documented.

The primary safety endpoint included device-related serious adverse events (SAEs) occurring within seven days of the index procedure or hospital discharge, whichever was later. Events were classified as primarily device or procedure related and adjudicated by an independent Clinical Events Committee (CEC). Only patients in whom a catheter was introduced during the index procedure were included in the analysis of the safety endpoint (safety population, see **Figure 1**).

Secondary Analyses

A CF ≥ 10 g was retrospectively considered optimal during creation of RF lesions.¹² This was based on data generated as this study was ongoing that demonstrated a high percentage of ineffective PV isolation (PVI) lesions using a low CF. Therefore, ablation results were evaluated for two complementary patient groups: patients with $\geq 90\%$ of their lesions created with a CF ≥ 10 g (optimal CF group) and patients with $< 90\%$ of the lesions created with a CF ≥ 10 g (non-optimal CF group).

The patient's quality of life (QoL) was assessed at baseline and 12 months follow-up using the AFEQT questionnaire (AF Effect on QualiTY of life). The questionnaire enables evaluation of the overall score and three sub-areas, which are symptoms, daily activities and treatment concerns. Comparisons were made between both catheter types and between patients with a successful outcome versus an ablation failure.

The use of the 3D mapping system was linked to the subject's randomization. To evaluate the potential for operator bias due to experience related to the mapping system,

operators were asked anonymously to rate their prior experience with the EnSite NavX and CARTO 3D mapping systems. Rating was relative to EnSite NavX system experience, with 50% showing equal experience with both 3D mapping systems. Ratings were averaged per center and weighted by the number of patients treated by the center. The impact of prior experience with either 3D mapping systems on the protocol-defined success rate was evaluated.

Statistical Methods

Statistical analysis was performed to evaluate the non-inferiority of the study device compared to the control with respect to the primary safety and effectiveness endpoints. Non-inferiority margins of 9% and -15% were used for the analysis of the safety and the effectiveness endpoints. One-sided 95% confidence limits for the risk difference of TactiCath minus Control (lower limit for effectiveness, upper limit for safety) were calculated based on Farrington-Manning approach. Also, p-values for non-inferiority testing using Farrington-Manning-test were provided. Non-inferiority margins were agreed with the US Food and Drug Administration (FDA) prior to the study.

For the sample size calculations related to the primary effectiveness endpoint, treatment success rate in both treatment groups was assumed to be around 55% based on previous studies.^{3,}

¹⁸ A total of 274 patients provides 80% power to show non-inferiority using a non-inferiority margin of -15% and a one sided 5% significance level. For the primary safety endpoint, the incidence of primary SAEs within 7 days of the index procedure was expected to be 9% in both treatment arms. A total of 274 patients provides 83% power to show non-inferiority using a non-inferiority margin of 9% and a one sided 5% significance level. Sample size was increased by 10% to account for attrition, resulting in a total of 300 randomized subjects. To investigate potential treatment differences across study centers, a logistic regression analysis for treatment

success was performed including treatment, center and treatment-by-center interaction as explanatory variables.

Categorical variables are expressed as counts and percentages and were compared with Fisher's exact test. Continuous variables are reported as means $\pm$ standard deviations or medians when relevant, and were compared using the Mann-Whitney U test. One-sided tests were used in comparisons involving the optimal CF group. Two-sided tests were used otherwise. A p-value below 0.05 was considered to be statistically significant.

Results

A total of 317 patients were enrolled in this study of which 17 were roll-in patients (**Figure 1**). Of the 300 patients who were randomized, 5 patients were excluded from analysis as no ablation was performed. The remaining 295 patients were included in the analysis of the primary safety endpoint (safety population). Patients in whom PV isolation was attempted and who had no major protocol deviations (N=280) were analyzed for the effectiveness endpoints (146 and 134 patients were treated with the CF and control catheters, respectively).

Patient Characteristics

Baseline patient demographics of the 295 patients comprising the safety population are summarized in **Table 1**. The mean age of all patients was 60.3 ± 10.1 years and 64.7% were males. The mean left atrial diameter and left ventricular ejection fraction was 39.6 ± 5.3 mm and $62.5 \pm 6.7\%$ respectively. A history of structural heart disease was present in 5.8% of patients. There were no significant differences between the patients assigned to each treatment arm.

Procedural Characteristics

The median fluoroscopy and radiofrequency ablation times were 27.0 vs. 23.0 min ($p=0.044$) and 46.5 vs. 53.0 ($p=0.018$) for the CF group and Controls, respectively. During the 30 min waiting

period 12.1% of lesions sets, had PV reconnections; 10.4% in CF group vs. 13.8% in Controls (p=0.206). Ultimately, all PVs were electrically isolated in both treatment groups.

Effectiveness Endpoint

Acute ablation success was achieved in 100% of patients in both treatment groups as all targeted PVs were successfully electrically isolated. Primary effectiveness endpoint at 12 months was achieved in 67.8% and 69.4% for the CF and Control groups, respectively (absolute difference - 1.6%, lower limit of 95% CI -10.7%, p=0.0073 for non-inferiority) which was above the pre-defined non-inferiority margin of -15% (**Figure 2A**). Study endpoint was also met even under 'worst case' conditions of the mITT population (290 patients). Clinically-relevant success at 12 months was achieved in 78.1% and 80.6% of the patients treated with CF catheter and the Control, respectively (p=0.659) (**Figure 2B**). To allow comparison with historical studies including the recently reported SMART-AF study, an additional analysis was retrospectively performed applying the definition of a protocol-defined success but allowing previously failed AAD during the effectiveness period ('on drug' analysis).¹³ Using this approach, success rates of 72.6% and 73.9% were achieved with the study device and the Control, respectively. For the primary endpoint, there was no treatment difference detected across study centers (p=0.20 for treatment-by-center interaction term).

Thirty three subjects (11.8%) underwent a repeat ablation procedure during the assessment period after blanking (16 CF, 17 Control). One subject in the Control arm required a third procedure. There were 12 subjects with repeat ablation during the blanking period (6 in each treatment arm).

Optimal vs. Non-Optimal Contact Force

On retrospective analysis, data suggested that when $\geq 90\%$ of the lesions were created with a CF $\geq$

10 g there appeared to be more durable PV Isolation. During PV isolation with the CF catheter, an optimal CF ($\geq 90\%$ of the lesions created with a CF ≥ 10 g) was achieved in 57.2% of the patients. Ablation with optimal CF resulted in significantly higher success rates compared with results from PV isolation with non-optimal CF. The protocol-defined success was 75.9% for optimal CF vs. 58.1% for non-optimal CF groups ($p=0.018$). The clinically relevant success rate was 85.5% vs. 67.7% for those with optimal CF compared to those with non-optimal CF ($p=0.009$) (**Figure 2C & 2D**). There were no statistically significant differences between optimal and non-optimal CF patients in terms of the baseline characteristics as reported in **Table 1**. Using the ‘on drug’ analysis, the success rate was 79.5% in the optimal CF group vs. 62.9% in the non-optimal CF group ($p=0.022$). There was a trend towards a lower rate of repeat ablations in the optimal CF group compared to the non-optimal CF group (7.2% vs. 16.1% $p=0.078$) and to Controls (7.2% vs. 12.7%, $p=0.148$).

Among the 36 operators who used the study device, 10 operators achieved an optimal CF in all of their patients, whereas 13 operators did not achieve optimal CF in any of their patients (**Figure 3A**). Operators that realized an optimal CF in $>80\%$ of their patients achieved a significantly higher protocol-defined success rate than others (79.1% vs. 58.2%, $p=0.008$, **Figure 3B**). Contact force was significantly different (26.5 vs. 19.2g, $p < 0.001$) between the two groups (**Figure 3C**). Interestingly, the mean power in the group that used optimal CF was also lower (28.0 vs. 26.9W, $p<0.001$). Although this difference is slight, it suggests that optimal CF may allow for less power.

Impact of Experience with 3D Mapping System

Prior to entry in the study, sites were required to demonstrate comparable proficiency with both of the 3D mapping systems used in the study. Surveys indicated that sites’ experience with the

EnSite™ NavX™ system was proportionately lower ($42.0 \pm 23.2\%$) showing a trend towards more experience with CARTO ($58.0 \pm 23.2\%$), which was used in all control catheter cases. Experience ratings were divided in four categories with approximately similar numbers of patients in each category (**Figure 4**). Operators with a far below average experience with the EnSite system (lowest group in **Figure 4**) achieved a significantly lower protocol-defined success rate with the study device than those with average or above average experience with the EnSite system ($p=0.002$). For the least experienced EnSite system operators, treatment success occurred in 47.5% of patients, compared to 75.7% ($p=0.019$) in those with equivalent experience and 80.0% ($p=0.004$) in those with more EnSite system experience.

There were a total of 77 patients (40 CF, 37 Control) treated by operators with <25% EnSite experience. When these patients were excluded from the primary effectiveness analysis, the success increased from 67.8% to 76.0% in the CF group vs. from 69.4% to 69.8% for Controls. Of course, due to the smaller sample size the increased success observed in the CF group when the less experienced EnSite operators were excluded did not reach statistical significance.

Quality of Life

Quality of life changes from baseline to 12 months are presented in **Figure 5**. Significant QoL improvements were observed in both arms with a non-significant trend for the CF vs. Control. Overall, patients with a clinically relevant success showed a significantly higher improvement than those with a failure ($p=0.002$). This effect was stronger in the Control arm compared to the CF arm ($p=0.008$ vs. $p=0.051$). For the sub-components, these effects were significant for the control device (except the ‘Symptoms’ component) and non-significant for the CF arm.

Safety Endpoint

There was no difference between the CF and Control groups with respect to overall serious

adverse events (device or procedure related) which occurred in 7.2% and 9.1% of patients, respectively (Table 2). The primary safety endpoint, which consisted of primary device-related SAEs, occurred in 1.97% and 1.40% of patients in the CF group and Controls, respectively (absolute difference 0.57%, upper limit of one-sided 95% CI 3.61%, $p=0.0004$ for non-inferiority). This is below the pre-defined non-inferiority margin of 9% and fulfilled non-inferiority criteria for safety. In the CF and Control groups, three and two primary device-related SAEs occurred, respectively. There were no deaths, no strokes, transient ischemic attacks, or atrioesophageal fistulas in either group.

Discussion

In this manuscript, we report the first prospective, multicenter, randomized controlled trial comparing a real-time CF sensing irrigated RF ablation catheter (TactiCath) to a standard irrigated RF ablation catheter (ThermoCool Navistar). In the trial, 300 patients with drug refractory paroxysmal AF were randomized in a 1:1 manner to undergo pulmonary vein isolation using either a force sensing or non-force sensing catheter. Both the primary effectiveness and primary safety endpoints for non-inferiority of CF catheter vs. Control were met. These endpoints fulfilled non-inferiority criteria even when using the more recent and stringent one-sided CI margins of 97.5% for the risk difference (-12.4% for primary effectiveness and 4.42% for primary safety endpoint).

Ablation Efficacy

Acute procedural success was achieved in all patients with 100% of targeted PVs being successfully isolated during the index procedure with both catheters. Following the blanking period, the 12-month drug-free rate of freedom from atrial arrhythmias was 67.8% for the CF

group vs. 69.4% for Control ($p=0.0073$ for non-inferiority). Several factors may have contributed to the failure of the CF catheter to demonstrate superior efficacy compared to the control catheter. Firstly, this study was designed to demonstrate non-inferiority rather than superiority and a much larger sample size would have been necessary to demonstrate this. Two other important factors include i) lack of optimal CF delivery during the study, and ii) operator effect.

Optimal Contact Force

The study protocol did not specify CF guidelines for ablation. However, during the conduct of the Toccstar study, clinical results from other concurrent CF studies became available and were shared with operators. These studies provided optimal CF guidelines which were then incorporated by some operators during this study. Therefore, not all patients in this study underwent ablation according to these optimal CF parameters and this may have partially accounted for lack of superiority of the CF catheter in this study. That is, one cannot reasonably expect a tool to enhance a procedure if the tool was not being used optimally.

Optimal CF parameters were based on several studies including EFFICAS I where operators were blinded to CF information while performing PV isolation. Results showed that PV reconnections were more frequent in segments with lower CF and FTI following an invasive mapping procedure at 3 months¹⁶. Based on data gleaned from this study, EFFICAS II was conducted which provided un-blinded operators the following CF guidelines for ablation (minimum CF ≥ 10 g, target CF ≥ 20 g, minimum FTI > 400 gs). Comparing the PV reconnection rate at 3 months to that of the EFFICAS I patients, PV reconnections were decreased by 90% when CF guidelines were met (unpublished data). The TOCCATA study demonstrated that when PV isolation was performed with an average CF < 10 g, AF recurrence was 100%. When the

average CF was >20 g, AF recurrence was reduced to 20% ($p=0.01$).¹² Additionally, this study also demonstrated that intermittent contact is more prevalent at low CF which has implications on lesion quality. Together, these studies demonstrated that procedural success is highly dependent upon achieving a minimum CF force during ablation and avoiding CF instability caused by intermittent contact.

Based on these observations, a retrospective analysis was performed in this study where patients who had undergone ablation with optimal CF were compared with those who did not. In this analysis, optimal CF was defined as $\geq 90\%$ of the lesions created with a CF ≥ 10 g. When the CF arm was divided into sub-groups based on optimal CF vs. non-optimal CF, primary effectiveness success occurred in 75.9% vs. 58.1% of subjects ($p=0.018$), and clinically relevant treatment success rate in 85.5% vs. 67.7% ($p=0.009$). This significant improvement in efficacy suggests an important finding related to the value of maintaining stable CF during RF delivery; more lesions delivered at low CF significantly increases the risk for arrhythmia recurrence. Previous guidance and training has justifiably focused on the importance of positional stability of the catheter tip to achieve good lesions. However, positional stability does not guarantee stable contact force, especially when the total force is below 10g and there is intermittent contact. The TOCCATA study has shown that stable CF is critically important to ensure treatment effectiveness. Since stable and optimal CF is associated with positional stability, operators should critically focus their attention on measures of CF stability during lesion creation.

Operator Effect

In this study, the CF achieved by operators was highly variable. In the CF arm, twelve operators achieved an optimal CF in their patients and a corresponding high rate of treatment success (79.1%). Additionally, subjects treated by these operators required lower RF power while

achieving a better outcome. Finally, the rate of serious adverse events was somewhat lower in this group (6.0 vs. 6.3%). These findings suggest that the use of CF may facilitate diligence in positioning the catheter prior to ablation and maintaining stable contact during RF, thus contributing to improved safety and efficacy.

Measures of procedural efficiency, such as the rate of acute reconnection, revealed that operators successfully maintained a low rate of acute reconnection throughout the trial at approximately 11% of PVs in the CF arm. In contrast, the Control group experienced a much higher rate of acute reconnection initially (18%) which was subsequently reduced to 9% at the end of trial. Although the acute reconnection rate is not uniquely correlated with treatment success, it appears that operators adapted their use of the control device based on their experience with the CF catheter. Indeed, most operators anecdotally claimed this learning effect.

Relative equivalence in study outcomes between the two catheters may partially be accounted for by the relative operator experience with each mapping system. The primary goal of the TOCCASTAR trial was to compare the CF and control catheters. As each catheter was linked to a different mapping system (Control with CARTO, and the CF catheter with EnSite); thus, this study was not a pure comparison of CF sensing vs. no CF sensing. The majority of AF ablations have historically been performed with the CARTO system; indeed, operators in this study tended to have less experience with the EnSite system. The primary effectiveness endpoint was much higher when operators had equivalent (75.7%, $p=0.019$) or more experience (80.0%, $p=0.004$) with the EnSite system compared to those with the least experience (47.5%) with this system. When the patients that were ablated by operators with the least EnSite experience (<25%) were excluded, primary effectiveness increased from 67.8% to 76.0%. Of course, this numerical difference did not reach statistical significance due to the smaller sample size.

Although both catheters were found to be equivalent, this study demonstrated that efficacy may have been improved with the CF catheter if operators delivered optimal CF and were more experienced with the EnSite mapping system.

Safety

The safety profile of the CF catheter was similar to Control. Both study arms experienced low rates of primary (device-related) safety events, which occurred in 1.97% of patients in the CF group vs. 1.40% in the Control group ($p=0.0004$ for non-inferiority). The overall serious adverse event rates (device or procedure related) were also similar between the CF (7.2%) and Control groups (9.1%). Of those, there was one procedure and one device related cardiac tamponade in each group (1.3% for CF group and 1.4% for Control). This rate is in line with other studies that report a cardiac tamponade rate of approximately 0.9–1.3%.^{3, 19, 20} The recently published SMART-AF trial with another CF sensing ablation catheter reported a higher cardiac tamponade rate of 2.5%.¹³ The reason for the differences in this complication rate between the two catheters is not clear but may be due to differences in catheter design and stiffness and CF delivery differences. One subject in the Control group had PV stenosis. There were no manifest cerebral thromboembolic events or atriopharyngeal fistulas in either group.

The safety of the CF catheter compared favorably with prior studies.^{3, 4, 13} However, it is known that safety events can be related to excessive RF power or CF. In particular, prior pre-clinical investigations suggest that at power >30W or CF>30g the incidence of steam pops increases.^{14, 17, 21} While there was no specific recommendation in the Toccastar protocol as to the force or power that should be used, the manufacturer recommends that power be limited to 30W and CF to 30g in AF ablation based largely on prior published data. Most users in this study were generally experts with experience of several hundreds of AF ablations. However, it is possible

that use of optimal CF (in particular keeping CF <30g) may reduce complications among less experienced users.

Conclusions

This prospective, multicenter, randomized controlled trial demonstrated that the CF sensing ablation catheter was non-inferior to the standard non-force sensing catheter with respect to acute and chronic efficacy and safety. The trial also demonstrated that operator effects related to adherence to optimal CF guidelines and the use of a 3D mapping system may positively influence overall outcomes of AF ablation with the CF catheter. The use of optimal CF (>10g) was associated with higher success.



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Table 1. Demographics of all patients in the safety population.

	CF Catheter n = 152	Control n = 143	Total n = 295	p value
Age (years)	59.6±9.3	61.0±10.8	60.3±10.1	0.171
Gender (male)	100 (65.8%)	91 (63.6%)	191 (64.7%)	0.716
Coronary artery disease	10 (6.6%)	16 (11.2%)	26 (8.8%)	0.217
Heart failure	3 (2.0%)	3 (2.1%)	6 (2.0%)	1
Hypertension	75 (49.3%)	69 (48.3%)	144 (48.8%)	0.907
Diabetes	16 (10.5%)	17 (11.9%)	33 (11.2%)	0.717
Stroke or TIA	5 (3.3%)	12 (8.4%)	17 (5.8%)	0.080
Structural heart disease	6 (3.9%)	6 (4.2%)	12 (4.1%)	1
AADs at baseline	132 (86.8%)	125 (87.4%)	257 (87.1%)	1
Left atrial diameter (mm)	[n = 135] 39.9±5.9	[n = 125] 39.3±4.5	[n = 260] 39.6±5.2	0.176
LVEF (%)	[n = 136] 62.4±7.1	[n = 127] 62.4±6.2	[n = 263] 62.4±6.7	0.903

AAD = anti-arrhythmic drug, LVEF = left ventricular ejection fraction, TIA = transient ischemic attack. Values are mean ± standard deviation or n (%).

Table 2. Serious adverse events (SAE).

Category of SAEs	CF Catheter n = 152	Control n = 143
Device-related SAEs (primary)	3 (1.97%)	2 (1.40%)
Cardiac tamponade / perforation	1 (0.66%)	1 (0.70%)
Pericarditis	2 (1.32%)	0 (0.0%)
Pulmonary vein stenosis	0 (0.0%)	1 (0.70%)
Procedure-related SAEs	10 (6.58%)	11 (7.69%)
Cardiac tamponade / perforation	1 (0.66%)	1 (0.70%)
Pulmonary edema	2 (1.32%)	2 (1.40%)
Vascular access complications	3 (1.97%)	3 (2.10%)
Hospitalizations (initial or prolonged)	4 (2.63%)	5 (3.50%)
Total of device- or procedure-related SAEs	11 (7.24%)*	13 (9.09%)

*2 patients in the CF group had device- and procedure-related events

Figure Legends:

Figure 1. Flow-chart of patient enrollment and randomization as well as the populations included in various analyses.

Figure 2. Kaplan-Meier survival curves with respect to protocol-defined and clinically relevant success. (A) Primary effectiveness success for CF (study device) and control catheters (control device). (B) Clinically relevant success for the two groups. (C) Primary effectiveness success for optimal and non-optimal CF groups. (D) Clinically relevant success for optimal and non-optimal CF groups.

Figure 3. Analysis of optimal versus non-optimal contact force by operator. (A) Operator ranking based on the percentage of patients in whom an optimal CF was achieved (36 operators).

The dashed vertical line separates the operators who achieved optimal CF in $\leq 80\%$ of their patients (white bars: 24 operators, 13 with no optimally treated patients) from the ones who achieved optimal CF in $>80\%$ of their patients (black bars: 12 operators, 10 with all patients treated optimally). (B) Treatment success achieved by operators with optimal CF in $>80\%$ of their patients procedures compared to other operators. (C) Mean CF in at least and RF power for operators with optimal CF in $>80\%$ of the patients compared with other operators. Both variables were significantly different between the two groups ($p < 0.001$). Subjects treated by operators using optimal CF required lower RF power while achieving a better outcome.

Figure 4. Primary effectiveness success rate with respect to experience with the 3D mapping system. The number of patients in each category is indicated. The horizontal lines denote the overall success rate for both arms.

Figure 5. Quality of life changes from baseline to 12-month follow-up (mean + standard deviation). (A) Changes for CF and Control arms, (B) based on clinically relevant success.

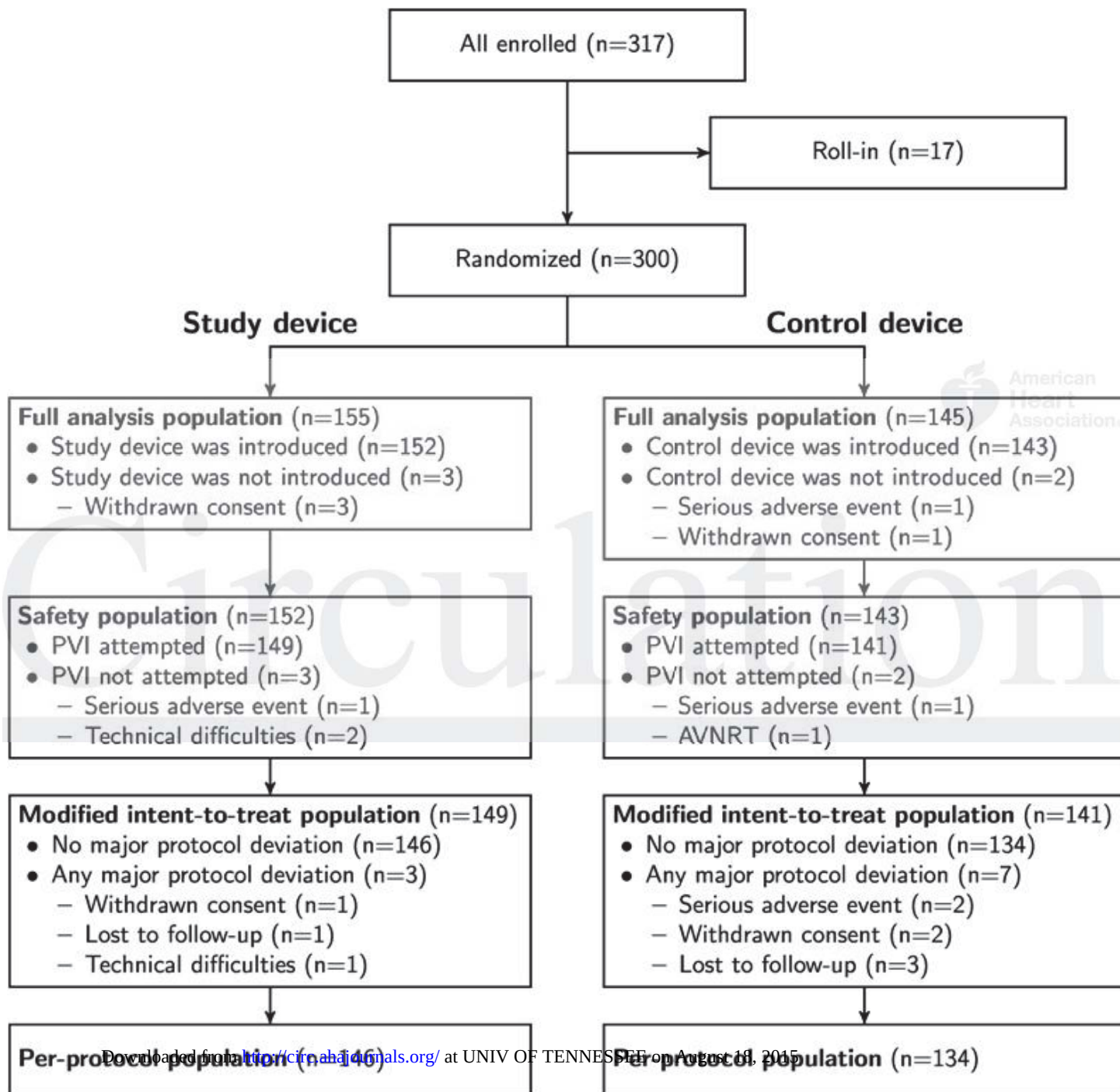


Figure 1

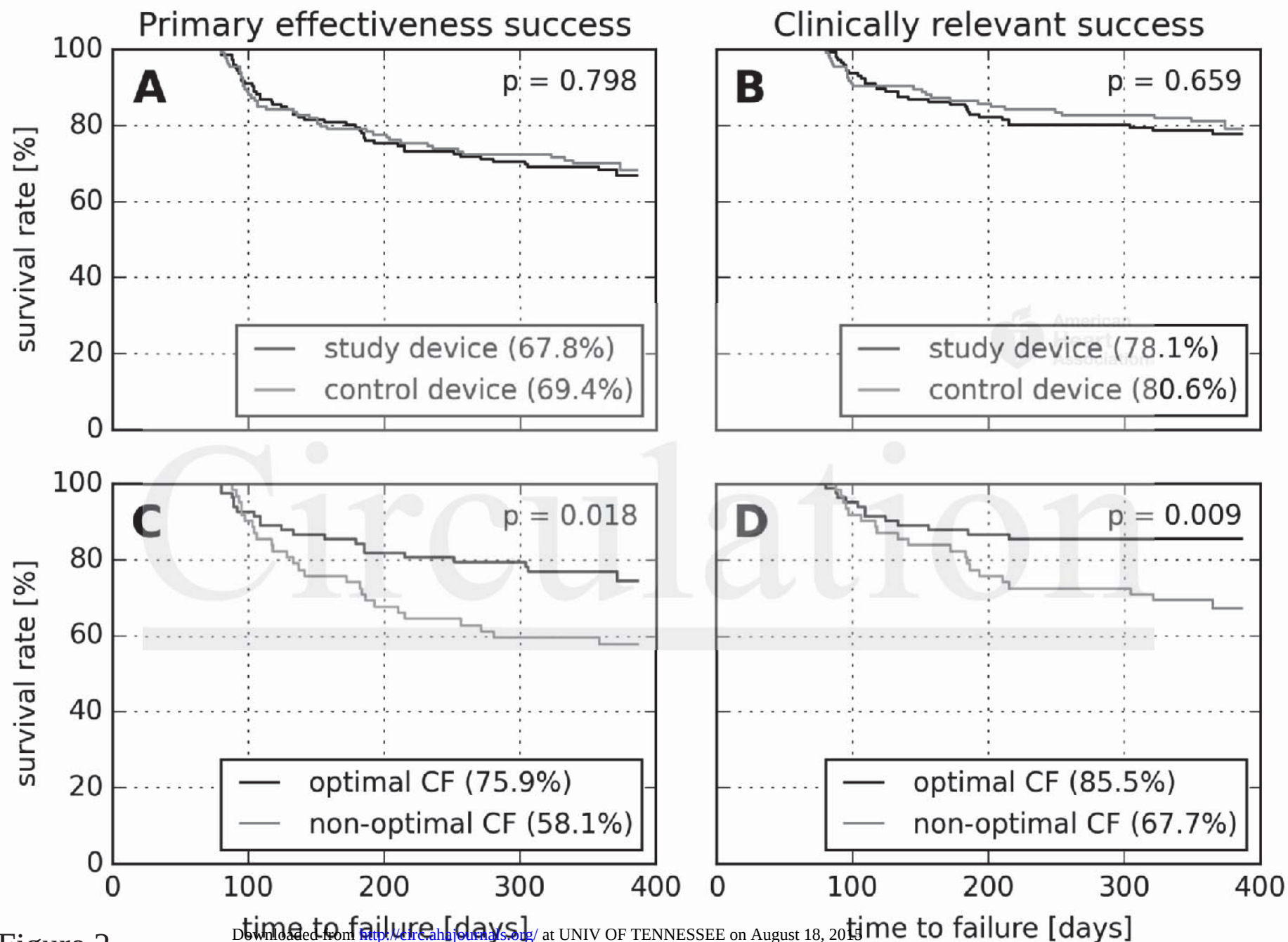
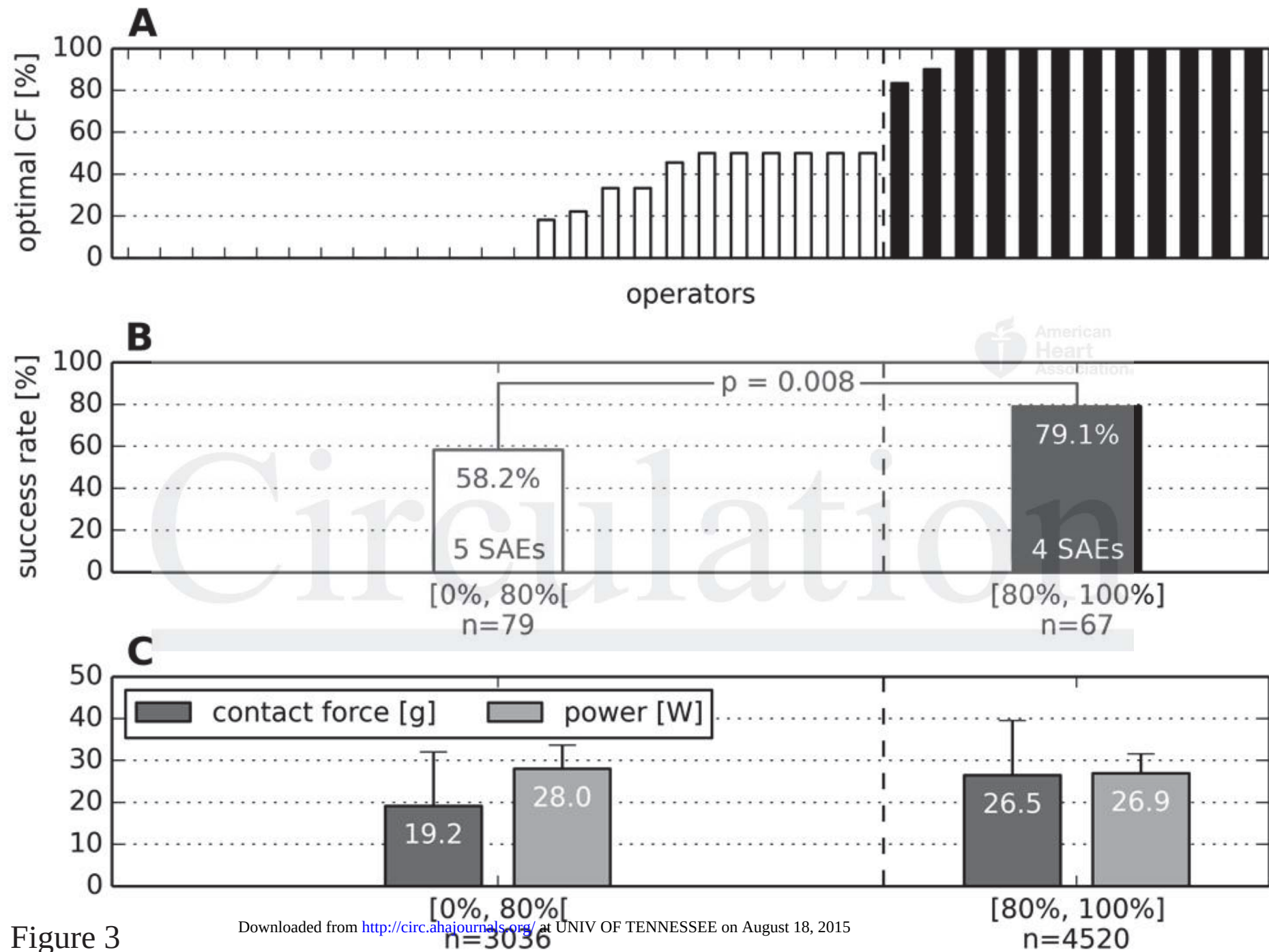


Figure 2



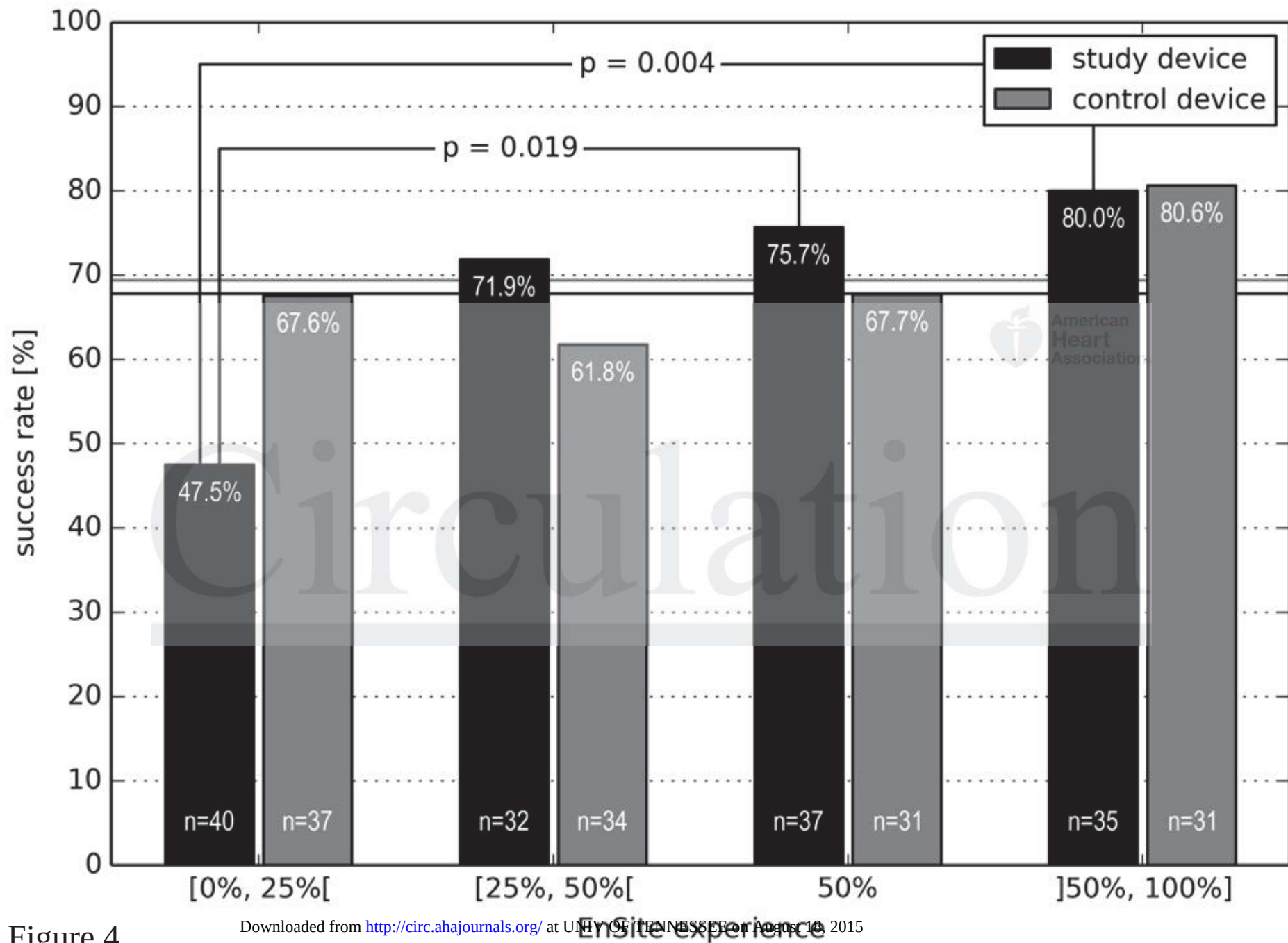


Figure 4

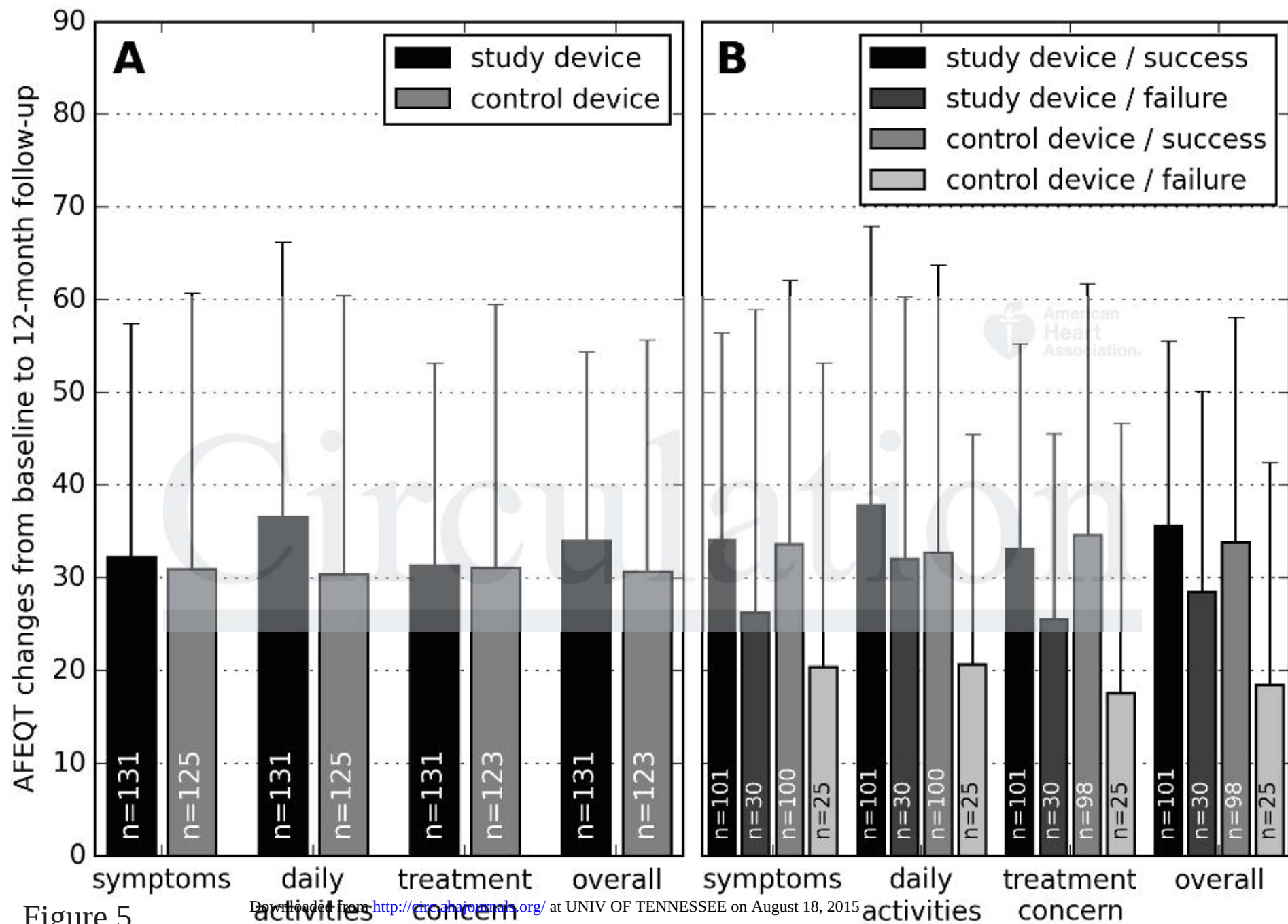


Figure 5

A Randomized Controlled Trial of the Safety and Effectiveness of a Contact Force Sensing Irrigated Catheter for Ablation of Paroxysmal Atrial Fibrillation: Results of the TOCCASTAR Study

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